

Solution Requirements

Date	30 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users can securely register and log in using username/email and password. Passwords are encrypted and sessions are authenticated.	High
2	Authorization levels	Two roles: Applicant (checks loan eligibility) and Admin (manages users, loan data, and ML model).	High
3	External interfaces	The system connects with the Machine Learning model and loan database to predict loan eligibility.	High
4	Transactions processing	The system accepts applicant details, validates them, processes them using the ML model, and predicts loan approval instantly.	High
5	Reporting	Displays loan eligibility result, approval probability, and allows downloading the prediction report as PDF.	Medium
6	Business rules, etc.	Prediction is generated only if mandatory details (income, credit score, employment status, loan amount, etc.) are provided.	High
7	Compliance to laws or regulations	Applicant information is securely stored and handled according to data privacy standards.	High
8	Other	Stores previous loan eligibility predictions for future reference and analysis.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Loan eligibility prediction should be generated quickly after user submission.	Prediction response time < 2 seconds.
2	Scalability	The system should support multiple users simultaneously.	Support 500+ concurrent users.
3	Security & Data Privacy	User credentials and financial details must be encrypted and protected.	Password hashing, HTTPS, SQL Injection protection.
4	Reliability & Availability	System should remain operational with minimal downtime.	99.9% availability.
5	Usability	User interface should be simple, responsive, and easy to understand.	Responsive design compatible with desktop and mobile devices.
6	Maintainability	The application should follow modular coding practices for easy updates.	Well-documented, reusable, and maintainable code.
7	Accuracy	The ML model should provide highly accurate loan eligibility predictions.	Prediction accuracy ≥ 90%.
8	Compatibility	The application should work across major browsers and operating systems.	Supports Chrome, Edge, Firefox, and mobile browsers.